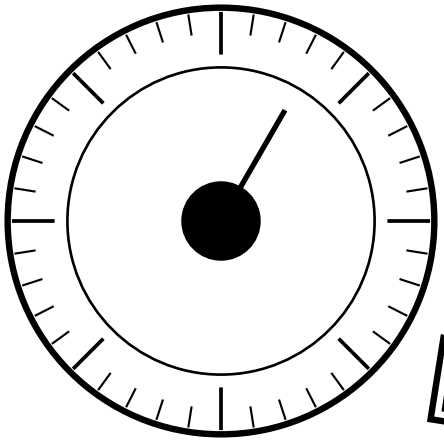
BONUS MISSION PACK

Ten extra missions for agents who finished the book,
or who just want more. Missions 81 to 90.
Hints, check codes and full solutions are inside.



FOR AGENTS ONLY

FUN BOOKS
PUBLISHER

LOGIC GRID | LEVEL ■■□ ADVANCED TYPICAL TIME: 14 MIN

SECTOR ATLAS SORT-OUT

Headquarters lost the roster file during last night's system update. These notes are the only backup copy you have left.

RULES Each agent matches exactly one item in every category, and no two agents share an item. Mark a box with X for no and O for yes. Earlier, later, lower and higher mean by any amount unless the clue says exactly.

		ROBOT				CODE NAME				FLOOR			
		Bolt	Gear	Rivet	Spark	Comet	Otter	Lynx	Orca	Floor 1	Floor 2	Floor 3	Floor 4
AGENT	Isaac												
	Maya												
	Sadie												
	Kiara												
FLOOR	Floor 1												
	Floor 2												
	Floor 3												
	Floor 4												
CODE NAME	Comet												
	Otter												
	Lynx												
	Orca												

1. Sadie works with the robot Spark.
2. The agent on Floor 1 is either Sadie or Maya.
3. The agent with the robot Gear works on a lower floor than Kiara.
4. The agent with the robot Gear is either Sadie or Isaac.
5. Isaac is not code-named Lynx.
6. The agent with the robot Bolt works on Floor 2.
7. The agent code-named Orca works on Floor 1.
8. The agent on Floor 2 is code-named Otter.

Which agent is code-named Lynx?

ANSWER _____ CHECK CODE (PAGE 12) _____

KEYWORD CIPHER | LEVEL ■■■ ELITE TYPICAL TIME: 21 MIN

CODE NAME: BROKEN BEACON

Your trainer, Agent Marcus, wrote this test message. Prove you can read it without any help.

RULES | The secret alphabet starts with the keyword, with repeated letters left out, then the rest of A to Z in order. Each plain letter sits above its secret letter.

INTERCEPTED MESSAGE

V	W	P	L	C		P	C	U		O	F	Y

W	A	J	T	U		L	W	A	D	I

U	D	B	C	P

DECODER KEY

Keyword: WALRUS

PLAIN	A	B	C	D	E	F	G	H	I	J	K	L	M
SECRET													

PLAIN	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
SECRET													

What is the first word of the message?

ANSWER | CHECK CODE (PAGE 12)

LOCK CODE | LEVEL ■■□ ADVANCED TYPICAL TIME: 10 MIN

THE RUSTY TUMBLER

The new air-quality sensor sits behind a dial with no instructions. Layla kept notes on every failed try.

RULES | Each clue is an earlier try at the code. Its sentence says how many of its digits are in the code and how many of those are in the right spot. Any other digit of that try is not in the code. The code never repeats a digit.

	TRY	WHAT THE LOCK SAYS
1	7 2 5	One digit is correct and in the right spot.
2	1 4 7	One digit is correct but in the wrong spot.
3	9 6 5	Two digits are correct: one in the right spot, one in the wrong spot.
4	0 9 2	Nothing is correct.



SPOT TRACKER

Cross out a digit at the top once it is out.
Put an X where a digit cannot go, a circle where it must.

DIGIT	0	1	2	3	4	5	6	7	8	9
1ST										
2ND										
3RD										

What is the 3-digit code?

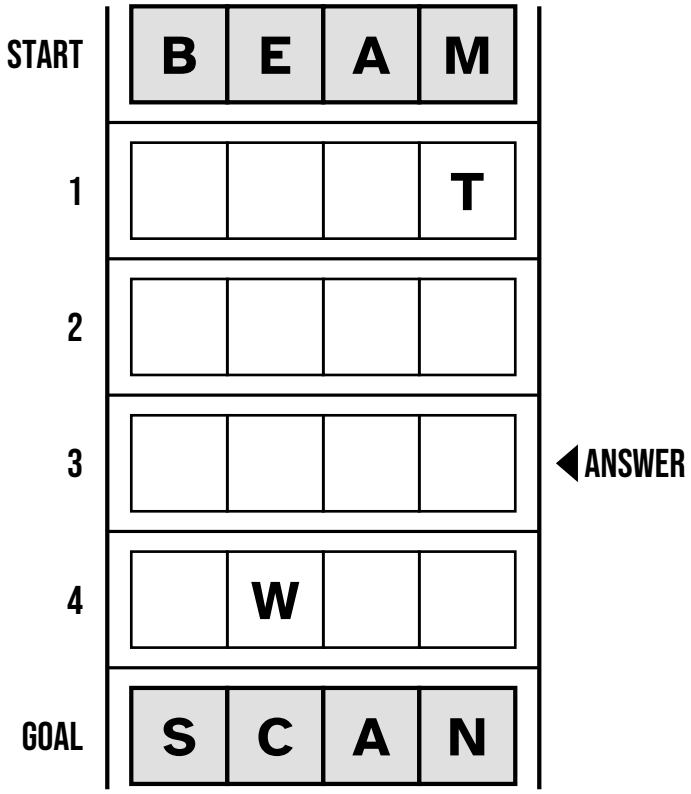
ANSWER _____ CHECK CODE (PAGE 12) _____

WORD LADDER | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

SIGNAL PATH: BEAM TO SCAN

A message drone dropped half its cargo. Use the printed letters to rebuild the climb from BEAM to SCAN.

RULES | Change exactly one letter at each step to climb from the top word to the bottom word. Every rung must be a real English word, and no word can appear twice. Printed letters are already in place.



WORK SPACE

What word sits on rung 3 of the ladder?

ANSWER _____ CHECK CODE (PAGE 12) _____

CRYPTOGRAM | LEVEL ■■□ ADVANCED TYPICAL TIME: 15 MIN

RADIO LOG: COBALT TIDE

The ocean lab sends its notes in code so no one can copy the results. Decode the newest note.

RULES | Each secret letter always stands for the same real letter, and no letter stands for itself. The shaded boxes give you three letters to start. Use the tracker to keep track.

INTERCEPTED MESSAGE

Shaded boxes are given.

G	A	H		K	V	W	D	U		M	D	R	H	W

C	V	U	B	K		D	Q	D	O	R		G	H	K	G
			K				G				.				

G	A	H		R	H	C		J	D	G	G	H	U	I
								B						

G	V	R	O	Q	A	G
				G			.

LETTER TRACKER

SECRET	A	B	C	D	E	F	G	H	I	J	K	L	M
REAL		K			—	—				B		—	

SECRET	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
REAL	—		—	G		—	—				—	—	—

A line in a box means that secret letter is not in the message.

What is the second word of the first sentence?

ANSWER _____ CHECK CODE (PAGE 12) _____

NONOGRAM | LEVEL ■■□ ADVANCED TYPICAL TIME: 12 MIN

THE SCRAMBLED SCREEN

An inventor hid a sketch inside this coded grid. Decode it to see what inspired her next project.

RULES Each number is a run of filled squares in that row or column, in order, with at least one empty square between runs. A 0 means the line stays empty. Fill in squares to reveal a picture, and put a dot in each square you know is empty.

	3	3	2	1 2	1 2	4 1	1 11	4 1	1 2	1 2	2	3	3
3													
1 1													
3													
7													
1													
1													
1													
1													
2 1 2													
2 1 2													
3 1 3													
3 1 3													
7													

What does the picture show? Write it in one word (6 letters).

ANSWER CHECK CODE (PAGE 12)

SYMBOL BALANCE | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

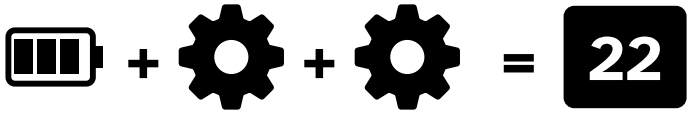
THE SPARK CHECK

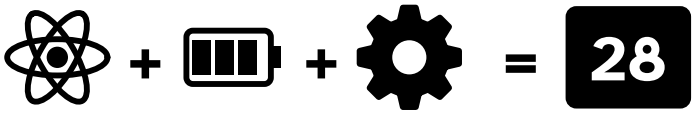
The control screen of the robot dog shows symbols instead of numbers after an update. Translate it back.

RULES | Each symbol stands for a whole number from 1 to 20. The same symbol always has the same value on every line. Every line balances: the left side equals the number on the right.

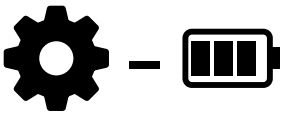
SYMBOL KEY

atom = ☐ battery = ☐ gear = ☐

1.  = 22

2.  = 28

3.  = 47

FIND () ×  = ☐

WORK SPACE

What is (gear – battery) × atom?

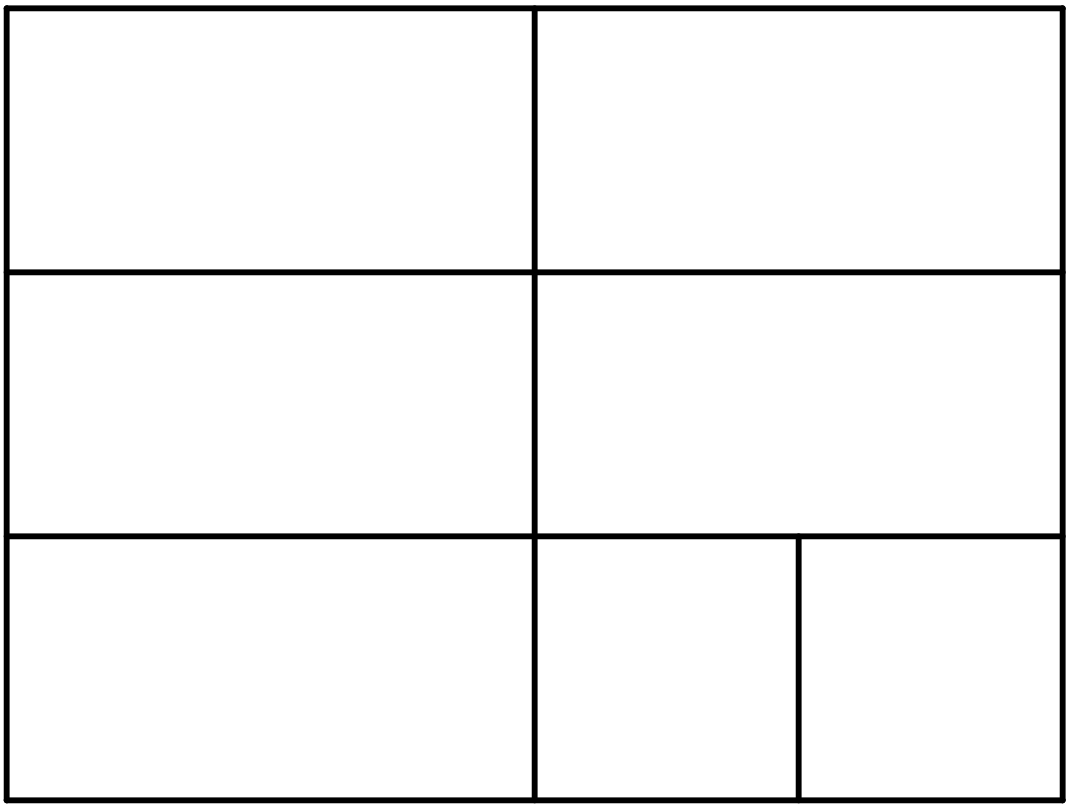
ANSWER ☐ CHECK CODE (PAGE 12) ☐

SHAPE COUNT | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

CROSSED PATTERN

A rival agent counted the rectangles in the telescope logo and got it wrong. Do better.

RULES | Count every shape of the kind asked, big or small, sitting flat or standing on a corner. Shapes can overlap and can be made by joining pieces. A square counts as a rectangle, but a rectangle does not count as a square.



TALLY BY SIZE ☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

One box per size. More sizes than boxes? Put two sizes in one box.

How many rectangles are in the figure, squares included? Write the number.

ANSWER ☐ CHECK CODE (PAGE 12) ☐

WORD MATH | LEVEL ■■□ ADVANCED TYPICAL TIME: 12 MIN

ADDING UP TO LIGHT

Your mentor wrote an addition on the board with a letter in place of each digit. Find the number LIGHT.

RULES Each letter stands for one digit from 0 to 9, and different letters stand for different digits. No number starts with 0. Add as usual, from right to left: when a column adds up to 10 or more, carry into the next column.

	W	A	T	C	H
+		S	I	G	N
	L	I	G	H	T

CODE KEY

A	C	G	H	I	L	N	S	T	W
7							8		

DIGITS USED (CROSS THEM OUT)

0	1	2	3	4	5	6	7	8	9

WORK SPACE

What number does LIGHT stand for?

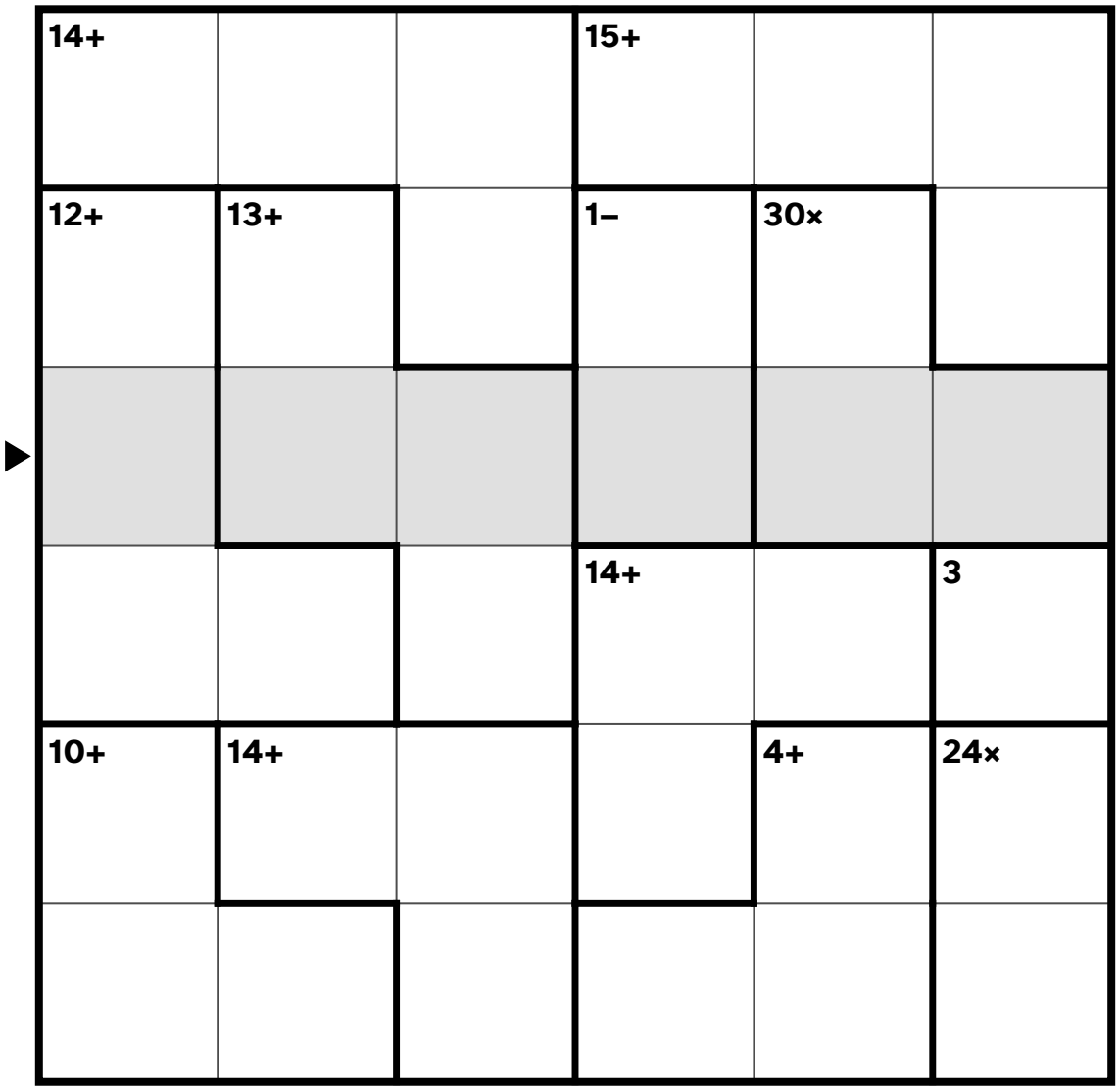
ANSWER _____ CHECK CODE (PAGE 12) _____

CALC CAGES | LEVEL ■■■ ELITE TYPICAL TIME: 20 MIN

THE COBALT PANEL

A training safe in the Field Map office waits for you. Solve the grid to open it.

RULES Each row and each column uses every number from 1 to the grid size exactly once. A thick outline is a cage: its numbers must make the target with the sign shown. For a minus or divide cage, start with the bigger number. A number can repeat inside a cage if the two are not in the same row or column. A one-square cage with no sign gives you its number.



Read the shaded row from left to right. What number do you get?

ANSWER _____ CHECK CODE (PAGE 12) _____

CHECK YOUR ANSWER

Turn your answer into a check code. Then look for your code in the list below.

If your code is in the list, your answer is almost certainly right. If it is not, check your work.

1. Write each letter as its number: A = 1, B = 2, C = 3 … Z = 26. A digit stays as it is.
2. Multiply the first by 1, the second by 2, the third by 3, and so on.
3. Add up the results. Then add ten times the mission number.

EXAMPLE · ANSWER "GEAR" IN MISSION 4

G	E	A	R
7 x 1 = 7	5 x 2 = 10	1 x 3 = 3	18 x 4 = 72

7 + 10 + 3 + 72 = 92

92 + 40 (mission 4 x 10) = 132

Check code: 132

CODES FOR MISSIONS 81 TO 90

Half of these codes are decoys.

819	**853**	**875**	**884**	**887**
914	**916**	**919**	**925**	**957**
965	**984**	**988**	**997**	**1015**
1029	**1075**	**1113**	**1144**	**1165**

Code not in the list? Check your arithmetic first. If the code is still missing, go back to the mission: one step went wrong somewhere. Take a hint if you want one.

HINT A

Read only the hint for your mission. Cover the rest with a sheet of paper.

DEPT 9 · BONUS FILES

M81 Mark every direct clue first. Then look hard at clue 2: it is the key, and the Either-Or technique unlocks it.

M82 Write WALRUS in the secret row, skipping any letter you already wrote, then add the rest of A to Z.

M83 Start with the clue that says nothing is correct, and cross out its digits everywhere.

M84 Start at the top: change one letter of BEAM so it matches the printed letters on rung 1.

M85 Write the given letters in every matching box first. Then look for the word that shows the most known letters, and think about its repeated letters.

M86 Start with the biggest clues, such as column 7 and row 11. Long runs in a short line leave little room to slide.

M87 Find two lines where one holds everything the other has, plus a little more.

M88 Sort the rectangles by how many small squares wide and tall they are.

M89 Start with the thousands column. Try each carry that can come out of the hundreds column.

M90 Start with the one-square cages, then look for cages with very few number sets.

HINT B

Try Hint A first. Hint B does the first step for you, never the whole mission.

DEPT 9 · BONUS FILES

M81 **[Direct Clue]** Clue 1 gives a match: Sadie works with the robot Spark. Mark the O and X the rest of its row and column.

M82 The secret alphabet begins W A L R U S B C D, and so the first secret letter V is W.

M83 Clue 4 (092): 0, 2 and 9 are out.

M84 First step done: rung 1 is BEAT.

M85 The secret word DQDOR is AGAIN, so D is A, O is I and R is N.

M86 Row 4 (clue 7) has nothing marked yet. Slide the run both ways, then fill square 7.

M87 Line 3 holds everything in line 2 plus an atom and a battery, so atom + battery = 47 – 28 = 19.

M88 There are 2 rectangles that are 1 by 1.

M89 **[Column Check]** Thousands column: 7 + 8 + the carry must end in I. The carry from the hundreds column can be 0 or 1. With no carry, I = 5, carrying 1. With a carry of 1, I = 6, carrying 1. So either way, the thousands column carries 1 and I is 5 or 6.

M90 The 14+ cage at row 5, column 2 can only use {2, 6, 6}, {3, 5, 6} and {4, 4, 6} and others.

Each solution names the technique it uses in bold. Learn the names: you will meet the same techniques again in harder missions.

MISSION 81 SECTOR ATLAS SORT-OUT

ANSWER: Kiara

Agent	Robot	Code Name	Floor
Isaac	Gear	Comet	Floor 3
Maya	Bolt	Otter	Floor 2
Sadie	Spark	Orca	Floor 1
Kiara	Rivet	Lynx	Floor 4

1. **[Direct Clue]** Clue 1 gives a match: Sadie works with the robot Spark. Mark the O and X the rest of its row and column. Mark every other clue that names a match or a non-match the same way.
2. **[Either-Or]** Clue 2 leaves only two agents for the agent on Floor 1, so every other box in that row gets an X.
3. **[Order Bounds]** Clue 3 sets an order, so the agent with the robot Gear does not work on Floor 4; Kiara does not work on Floor 2.
4. **[Last Box]** All the other code names are crossed out for this one, so Kiara is code-named Lynx.
5. Finish the grid with the same moves. The answer is Kiara.

MISSION 82 CODE NAME: BROKEN BEACON

ANSWER: WATCH

V	W	P	L	C		P	C	U		O	F	Y		
W	**A**	**T**	**C**	**H**		T	H	E		S	K	Y		
W	A	J	T	U		L	W	A	D	I				
A	B	O	V	E		C	A	B	I	N				
U	D	B	C	P										
E	I	G	H	T										

Secret alphabet: WALRUSBCDEFGHIJKMNOPQTVXYZ.
Answer: WATCH.

1. **[Build the Key Alphabet]** WALRUS without repeats is WALRUS; the unused letters follow, so the secret alphabet starts WALRUSBCDE.
2. **[Reverse Lookup]** Find each secret letter in the secret row and read the plain letter above it: V sits under W.
3. **[Word Check]** The message reads: WATCH THE SKY ABOVE CABIN EIGHT. Every word is a real word, so the first word is WATCH.

MISSION 83 THE RUSTY TUMBLER

ANSWER: 615

1 7 2 5 1 right spot

2 1 4 7 1 wrong spot

3 9 6 5 1 right spot, 1 wrong spot

4 0 9 2 nothing correct

CODE **6 1 5**

1. **[Nothing Is Correct]** Clue 4 (092): 0, 2 and 9 are out. **[Count the Digits]** Clue 3 (965): 5 and 6 are in the code.

MISSION 83 (CONTINUED)

2. **[Count the Digits]** Clue 1 (725): 7 is out. **[Right Spot]** Clue 1 (725): 5 goes in the third spot.
3. **[Wrong Spot]** Clue 2 (147): 1 is not in the first spot; 4 is not in the second spot.
4. **[Place Count]** Clue 3 (965): 6 goes in the first spot. **[Only Spot Left]** 4 has no spot left, so it is out.
5. **[Count the Digits]** Clue 2 (147): 1 goes in the second spot.
6. So the code is 615.

MISSION 84 SIGNAL PATH: BEAM TO SCAN

ANSWER: SWAT

START	**B**	**E**	**A**	**M**	
1	B	E	A	**T**	
2	S	E	A	T	
3	S	W	A	T	◀ ANSWER
4	S	**W**	A	N	
GOAL	**S**	**C**	**A**	**N**	

1. **[Step Down]** Change one letter of BEAM: only BEAT fits the printed letters on rung 1.
2. **[Step Up]** Work up from SCAN: only SWAN fits the printed letters on rung 4.
3. **[Count the Gap]** Next to SWAN, every other choice is too far from BEAT, so rung 3 is SWAT.
4. **[Squeeze Play]** Rung 2 must be one letter from BEAT and from SWAT, so it is SEAT.

MISSION 84 (CONTINUED)

5. The word on rung 3 is SWAT.

MISSION 85 RADIO LOG: COBALT TIDE

ANSWER: SOLAR

G	A	H	K	V	W	D	U	M	D	R	H	W
T	**H**	**E**	**S**	**O**	**L**	**A**	**R**	**P**	**A**	**N**	**E**	**L**

C	V	U	B	K	D	Q	D	O	R
W	**O**	**R**	**K**	**S**	**A**	**G**	**A**	**I**	**N** .

G	H	K	G	G	A	H	R	H	C
T	**E**	**S**	**T**	**T**	**H**	**E**	**N**	**E**	**W**

J	D	G	G	H	U	I	G	V	R	O	Q	A	G
B	**A**	**T**	**T**	**E**	**R**	**Y**	**T**	**O**	**N**	**I**	**G**	**H**	**T** .

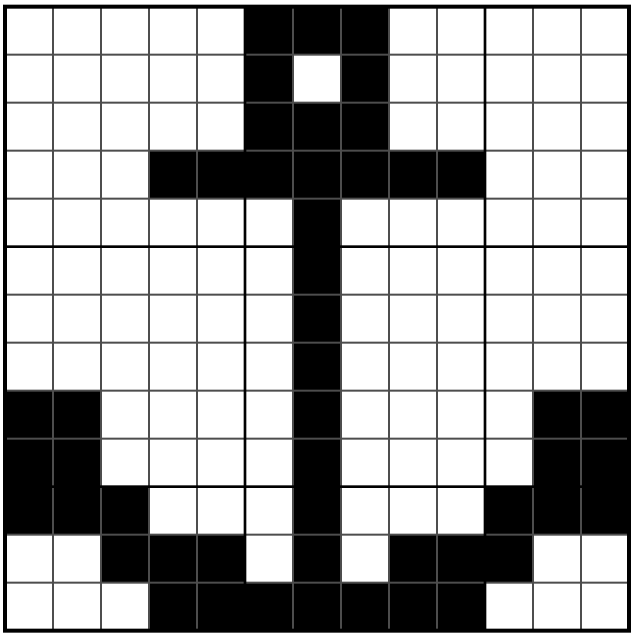
1. **[Given Letters]** Start with the letters you are given: B is K, J is B and Q is G. Write each one under every matching secret letter, and in the tracker.
2. **[Word Shape]** DQDOR now reads ?G???. Its length, its known letters and its pattern leave only AGAIN. So D is A, O is I and R is N.
3. **[Word Shape]** GVROQAG now reads ??NIG??. Its length, its known letters and its pattern leave only TONIGHT. So G is T, V is O and A is H.
4. **[Word Shape]** RHC now reads N??. Its length, its known letters and its pattern leave only NEW (NOT would need O, which is already taken; NOW would need O, which is already taken). So H is E and C is W.
5. Keep going with **[Word Shape]** and **[Fill the Gaps]**, one word at a time: CVUBK, which reads WO?K?, is WORKS, KVWDU, which reads SO?AR, is SOLAR and MDRHW, which reads ?ANEL, is PANEL, and the other words fill in the same way.

MISSION 85 (CONTINUED)

6. **[Read It Back]** The message reads: THE SOLAR PANEL WORKS AGAIN. TEST THE NEW BATTERY TONIGHT. Every word is real, so the second word of the first sentence is SOLAR.

MISSION 86 THE SCRAMBLED SCREEN

ANSWER: ANCHOR



1. **[Overlap]** Row 4 (clue 7) has nothing marked yet. Slide the run both ways, then fill square 7.
2. **[Match the Runs]** In row 11 (clue 3 1 3), match each marked square to its run. Then fill squares 3 and 11 and put a dot in squares 4-6 and 8-10.
3. **[Line Done]** Row 5 (clue 1) has all its filled squares, so put a dot in squares 1-6 and 8-13.
4. Keep switching between rows and columns. Each new square unlocks another line.
5. The finished picture shows an anchor.

MISSION 87 THE SPARK CHECK

ANSWER: 75

atom = 15 battery = 4 gear = 9

1 🔋 + ⚙ + ⚙ **= 22**

2 ⚛ + 🔋 + ⚙ **= 28**

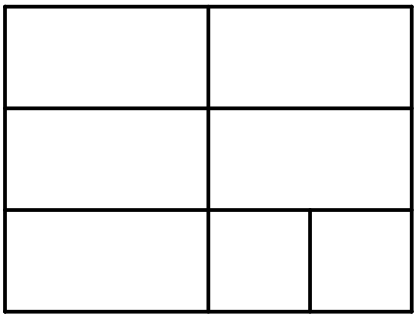
3 ⚛ + ⚛ + 🔋 + 🔋 + ⚙ **= 47**

? (⚙ **–** 🔋) **×** ⚛ **= 75**

1. **[Compare Scales]** Line 3 holds everything in line 2 plus an atom and a battery, so atom + battery = 47 – 28 = 19.
2. **[Compare Scales]** Line 2 holds everything in atom + battery = 19 plus a gear, so gear = 28 – 19 = 9.
3. **[Swap In]** Put gear = 9 into line 1: battery + 9 + 9 = 22, so the battery is 4.
4. **[Swap In]** Put battery = 4 and gear = 9 into line 2: atom + 4 + 9 = 28, so the atom is 15.
5. So (gear – battery) × atom = (9 – 4) × 15 = 75.

MISSION 88 CROSSED PATTERN

ANSWER: 21



Answer: 21 rectangles. Sort the rectangles by size, counting in small squares: 1 by 1: 2; 1 by 2: 6; 1 by 3: 1; 1 by 4: 3. The bigger ones: 2 by 2: 4; 2 by 3: 2; 2 by 4: 2; 3 by 4: 1.

1. **[Width x Height]** Sort the rectangles by size, counting in small squares: 1 by 1: 2; 1 by 2: 6; 1 by 3: 1; 1 by 4: 3.
2. **[Width x Height]** The bigger ones: 2 by 2: 4; 2 by 3: 2; 2 by 4: 2; 3 by 4: 1.
3. Add them all: 2 + 6 + 1 + 3 + 4 + 2 + 2 + 1 = 21 rectangles.

MISSION 89 ADDING UP TO LIGHT

ANSWER: 35610

	W 2	**A** 7	**T** 0	**C** 4	**H** 1
+		**S** 8	**I** 5	**G** 6	**N** 9
	L 3	**I** 5	**G** 6	**H** 1	**T** 0

A=7*, C=4, G=6, H=1, I=5, L=3, N=9, S=8*, T=0, W=2
* given

1. Write the given digits under their letters: A = 7, S = 8.

MISSION 89 (CONTINUED)

2. **[Column Check]** Thousands column: 7 + 8 + the carry must end in I. The carry from the hundreds column can be 0 or 1. With no carry, I = 5, carrying 1. With a carry of 1, I = 6, carrying 1. So either way, the thousands column carries 1 and I is 5 or 6.
3. **[Suppose and Crash]** Digits still possible for W: 1, 2, 3, 4, 5, 6, 9. If W = 1, the ten-thousands column gives L = 2, then the ones and tens columns cannot both add up. If W = 2, it fits. If W = 3, the ten-thousands column gives L = 4, then the ones, tens and hundreds columns cannot all add up. If W = 4, the ten-thousands column gives L = 5, then the thousands column needs I = 6 and a carry of 1 from the hundreds column, then the ones and tens columns cannot both add up. If W = 5, the ten-thousands column gives L = 6, then no digit for I makes the hundreds column add up. If W = 6, the ten-thousands column gives L = 7, but 7 is A. If W = 9, the ten-thousands column makes 10, so it carries 1, not 0. So W = 2.
4. **[Column Check]** Ten-thousands column: 2 + 1 (carried) = 3. So L = 3.
   [Suppose and Crash] Digits still possible for I: 5, 6. If I = 5, it fits. If I = 6, the thousands column needs a carry of 1 from the hundreds column, then the ones and tens columns cannot both add up. So I = 5.
   [Column Check] Ones column: H + N must end in T. The carry from the ones column can be 0 or 1. With no carry, no digits still free make the ones column add up. With a carry of 1, it fits. So the ones column carries 1.
   [Column Check] Thousands column: 7 + 8 + the carry must make 15, so the carry is 0 (7 + 8 + 0 = 15). So the hundreds column does not carry.

MISSION 89 (CONTINUED)

5. **[Carry Chain]** Hundreds column: T + 5 + the carry must end in G, with no carry. The carry from the tens column can be 0 or 1. With no carry, the tens and hundreds columns cannot both add up. With a carry of 1, the hundreds column needs G = 6 and T = 0. So G = 6, T = 0 and the tens column carries 1. **[Carry Chain]** Ones column: H + N must end in 0 and carry 1. Digits still possible for H: 1, 4, 9. If H = 1, N = 9, then the tens column gives C = 4. If H = 4, N = 6, but 6 is G. If H = 9, N = 1, then no digit for C makes the tens column add up. So C = 4, H = 1 and N = 9.
6. So LIGHT = 35610.

MISSION 90 THE COBALT PANEL

ANSWER: 462351

14+ 6	1	4	15+ 5	3	2
12+ 1	13+ 4	3	1− 2	30× 6	5
4	6	2	3	5	1
5	2	1	14+ 6	4	3 3
10+ 2	14+ 3	5	4	4+ 1	24× 6
3	5	6	1	2	4

1. **[Free Number]** The one-square cage at row 4, column 6 gives you a 3.
2. **[Cage Claim]** Every option for the 10+ cage at row 5, column 1 puts a 3 in row 6, so no other square there holds a 3.
3. **[Test and Reject]** For the 14+ cage at row 5, column 2, the set {3, 6, 5} leaves a number with nowhere to go, so cross it out.

MISSION 90 (CONTINUED)

4. **[Only Spot]** In row 5, a 2 fits only at row 5, column 1.
5. Keep going with Last Option and Only Spot until every square is full.
6. The shaded row reads 462351.